2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 9

Due Date: Due on 3/Dec/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student ID**. Write neatly on **A4-sized** paper (*staple if necessary*).
- **Late submissions or answers without steps will not be graded.**

---

1. For each of the following, find the transition matrix corresponding to the change of basis from $\gamma = \{\mathbf{u_1}, \mathbf{u_2}\}$ to $\beta = \{\mathbf{e_1}, \mathbf{e_2}\}$.

   (a) $\mathbf{u_1} = (1,1)^T, \mathbf{u_2} = (-1,1)^T$

   (b) $\mathbf{u_1} = (1,2)^T, \mathbf{u_2} = (2,5)^T$

   (c) What is the transition matrix from $\gamma_1 = \{(1,2)^T, (2,5)^T\}$ to $\gamma_2 = \{(1,1)^T, (-1,1)^T\}$?

2. Let $\mathcal{A} = \{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be bases for a vector space V, and suppose $\mathbf{a}_1 = 4\mathbf{b}_1 - \mathbf{b}_2$, $\mathbf{a}_2 = -\mathbf{b}_1 + \mathbf{b}_2 + \mathbf{b}_3$ and $\mathbf{a}_3 = \mathbf{b}_2 - 2\mathbf{b}_3$.

   (a) Find the transition matrix from $\mathcal{A}$ to $\mathcal{B}$, $[I]_{\mathcal{A}}^{\mathcal{B}}$.

   (b) Find $[\mathbf{x}]_{\mathcal{B}}$ for $\mathbf{x} = 3\mathbf{a}_1 + 4\mathbf{a}_2 + \mathbf{a}_3$.

3. Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 4 & -2 & 4 & -2 \\ -2 & 0 & 1 & 3 \end{bmatrix}$$

   (a) Find a basis for the row space and the column space of A, respectively.

   (b) Find a basis for the null space of A.

   (c) Find a basis for the null space of A^T.

   (d) Determine the nullity and the rank of A^T and verify the rank nullity theorem.

4. For each of the following A and $\mathbf{b}$, determine whether $\mathbf{b}$ is in the column space of A and state whether the system $A\mathbf{x} = \mathbf{b}$ is consistent. If consistent, further determine if it has the unique solution or infinitely many solutions without solving the system:

   (a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 4 \\ 8 \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

(c) $A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & -1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}$

(d) $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

5. Let A be a 4×4 matrix with reduced row echelon form given by

$$U = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

If

$$\mathbf{a}_1 = \begin{pmatrix} -3 \\ 5 \\ 2 \\ 1 \end{pmatrix} \quad \textit{and} \quad \mathbf{a}_2 = \begin{pmatrix} 4 \\ -3 \\ 7 \\ -1 \end{pmatrix},$$

determine $\mathbf{a}_3$ and $\mathbf{a}_4$.

6. Let A be a 4×5 matrix. If $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_4$ are linearly independent and

$$\mathbf{a}_3 = \mathbf{a}_1 + 2\mathbf{a}_2, \qquad \mathbf{a}_5 = 2\mathbf{a}_1 - \mathbf{a}_2 + 3\mathbf{a}_4$$

determine the reduced row echelon form of A.

7. Prove that a nonempty subset of a finite set of linearly independent vectors is linearly independent.